



DEPARTMENT OF Mathematics
Fall 2026-2027
COURSE SYLLABUS

I. GENERAL COURSE INFORMATION

Course Number	Math 203
Course Title	Mathematics for Social Sciences I
Number of Credits	3
Course Level	Undergraduate
Course Format	Lectures
Pre-requisite Courses	Math 100
Meeting Schedule	TR at 2:00 pm
Meeting Location	Nicely 323

II. COURSE INSTRUCTOR

Name	Role	Office Location	Office Hours	Extension	E-mail
Mona Al Batrouni	Instructor	Bliss113	MW - 10.30 am		Mab108@mail.aub.edu Ma224@aub.edu.lb

III. COURSE DESCRIPTION

Mathematical notations and basic notions; properties of real numbers; factoring polynomials; functions and their graphs; straight lines and parabolas and their equations; Gaussian elimination; exponential and logarithmic functions; limits and continuity; basic differential calculus. MATH 203 may not be taken for credit after a student has passed MATH 101, MATH 102 or MATH 201. Prerequisite: MATH 100. Every term.

IV. COURSE LEARNING OUTCOMES

Upon completion of the course, students will be able to:

- CLO1: Apply operations on polynomials.
- CLO2: Factor polynomials.
- CLO3: Solve first- and second-degree inequalities.
- CLO4: Solve linear inequalities and equations of absolute values;
- CLO5: Write the equation of a straight line;
- CLO6: Use the properties of the slopes to determine the equation of parallel lines and perpendicular lines;
- CLO7: Plot the graph of special functions in rectangle coordinates;
- CLO8: Apply the symmetry and reflection to functions;
- CLO9: Calculate the derivative of a function using the definition;
- CLO10: Apply the rules of differentiation of the sum, difference, product, and quotient;
- CLO11: Calculate $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow \pm\infty} f(x)$;
- CLO12: Apply the limit rules;
- CLO13: Test for continuity of a function at a point;
- CLO14: Use Gaussian elimination to solve a linear system of equations;
- CLO15: Apply the properties of the exponential and logarithmic functions;
- CLO16: Solve logarithmic and exponential equations.
- CLO17: Find the derivative of polynomials, logarithmic and exponential functions

V. PROGRAM LEARNING OUTCOMES ADDRESSED IN THE COURSE

I. PROGRAM LEARNING OUTCOMES ADDRESSED IN THE COURSE

Below is a detailed outline of how the specific PLO of the Mathematics department at AUB can be aligned with the course content.

PLO1: Communicate mathematical concepts orally and in writing.

Students write and provide orally clear explanation of mathematical procedures used in the essential topics such as basic calculus.

PLO2: Carry out significant mathematical computations in algebra, analysis, and geometry.

Students apply knowledge in solving inequalities, system of linear equations, logarithmic and exponential equations, calculation of limits and derivatives.

PLO3: State theorems and give examples where they are used.

Students state accurately properties of functions including piecewise-defined, exponential, and logarithmic functions.

PLO4: Develop clear, organized, and rigorous mathematical proofs.

Students ensure that all mathematical expressions, formulas, and calculations are accurately represented in the explanation of mathematical procedures

VI. COURSE RESOURCES AND REFERENCES

Text Book: Introductory Mathematics Analysis for Business, Economics, and the Life and Social Sciences by Ernest F. Haussler, Richard S. Paul, Richard J. Wood, Fourteenth Edition (E-book Pearson Edition)

VII. COURSE OUTLINE			
Week	Chapter/Topic	Sec.	Suggested problems
Aug.31 – Sep.4	Chapter 1	1.2	p.58 :12,14,18,20,22,24,26,30,33,34.
Sep.7– Sep.11	Applications and more Algebra	10.4 1.4 1.5	p.479: 2,6,10,11,12,14,16,18,22,24,26. p.65 :16,18,20,21,22,24,26,28,30,32,33,34,35,36. p.70: 4,6,8,12,13,14,16,17,20,22,26.
Sep.14– Sep.18	Chapter 2	2.1	p.90: 2,3,6,8,10,12,14,16,20,22,29,32,38.
Sep.21 – Sep.25	Functions and Graphs	2.2 3.3	p.95: 2,3,4,6,8,10,11,12,19,20,22. p.151: 2, 6, 9,11,15,16,20,24,26.
Sep. 28– Oct.2		2.5 2.7	P.110: 4, 6,10,12,14,16,20,22,27,30,32,34,36,38. p.120: 1, 2,4,6,7,9,10.
Exam I Wednesday September 30, 2026 at 6:00 pm			
Oct.5– Oct.9	Chapter 3 Lines, Parabolas and Systems Gaussian Elimination method	3.1 Extra	p.138: 2,4,8,10,11,13,18,20,21,22,24,26,32,34, 36,38,42,43,44,46,50,52,54,56,60,62. Check Moodle
Oct.12– Oct.16 Oct.19 – Oct. 23	Chapter 4 Exponential and Logarithmic Functions	4.1 4.2 4.3 4.4	p.186: 2, 4, 8,9,10. p.193: 2,4,6,8,17,20,24,25,26,28,30,32,34,35,38, 40, 42, 44, 46, 48,50,52,54. p.199: 1,4,6,10,12,14,15,17,18,20,22,24,28,32,34, 36, 38, 41, 42, 44,46,48,52. p.201 :2,3,4,6,7,8,9,12,14,20,25,27,30,32,34,35.
Oct.26 – Oct.30 Nov.2 – Nov.6	Chapter 10 Limits and continuity	10.1 10.2 10.3	p.459: 2,4,12,16,18,20,22,24,30,32,34,35,38,43. p.467 :6,7,10,13,16,17,20,22,23,26,28,30,33,34,38, 41,43,51,56,58. p.474: 2,5,10,11,12,20,21,25,30,32,34.
Exam II Wednesday November 4, 2026 at 6:00 pm			
Nov.9 – Nov.13 Nov.16–Nov.20	Chapter 11 Differentiation	11.1 11.2 11.4 11.5	p.490: 4, 12. p.498: 2,6,10,12,14,18,20,22,26,28,36,38,40,42,44, 46, 48,54,58,64,70,72,74. p.517: 2, 4, 12, 17,20,26,30,34,41,45. P.525 :2,5,6,10,14,18,20,26,28,32,34,36,40,42,44, 51,55,56.
Nov.23– Nov.27 Nov.30–Dec.4	Chapter 12 Additional Differentiation Topic	12.1 12.2 12.5 12.7	p.536 :2,4,5,10,12,14,15,20,22,26,30,32,36,38,40,44 p.541: 1,4,8,10,14,16,20,24,25,26,28,30. p.558: 2,4,7,8,10,12,14,16,17,19,21,22. p.565: 2, 4, 5, 7,12,17,18,20,21,22.
Final Exam			

VIII. GRADING CRITERIA

Type of Evaluation	Percentage
Exam 1	Max. (Exam 1, Exam 2) 30 %
Exam 2	Min. (Exam 1, Exam 2) 25%
Attendance	5%
Final Exam	40%
Total	100%

IX. COURSE POLICIES

- You are expected to attend **all** class meetings. If you are unable to attend class meetings, it is **your responsibility** to inform the instructor prior to the class meeting. Absences will be dealt with on a case-by-case basis.
“Students who miss more than one-fifth of the sessions(**with or without excuse**) of any course prior to the withdrawal deadline can be withdrawn from the course by the instructor.”
- The university policy regarding incomplete work, attendance and other issues will be followed. **Check the AUB catalogue for details.**
- Students are responsible for all the material presented in class meetings, **including announcements about course procedure**. Students who miss classes are responsible for any announcement made during their absences.
- The syllabus, some previous quizzes and some extra sheets will be posted on Moodle.
- Normally, we use curving that assigns grades to students relative to the class as a whole, based on their performance (Class evaluation, attendance, short quizzes, homework ...), students that **miss 5 or more class meetings will not** be put on the curve and **will not** get any raise on the final average at the end of the semester.
- Please refer to AUB Student Code of Conduct:
<https://www.aub.edu.lb/SAO/Documents/student%20code%20of%20conduct.pdf>
in particular section 1, which concerns academic misconduct including cheating, plagiarism, in-class disruption, and dishonesty. Please be aware that misconduct is vigorously prosecuted and that AUB has a zero-tolerance policy. Course policy is that **credible evidence of cheating will result in course failure**.
- Withdrawal:** Last day for withdrawal from courses, Friday Nov 20, 2026.
- Credit Hour Policy**
“This course follows AUB Credit Hour Policy for 3-credit courses. This course meets three *times* weekly for three academic hours per week.
- Final Examinations:**
Please note that students are expected to be on-campus and available to take a final examination for any of their courses at any time during the final examination period. For the current term, the final examination period runs between **Dec.9 and Dec.18,2026**. Students and professors will be notified of the final examination schedule being posted on the Office of the Registrar website as soon as feasible after the change of schedule period (drop and add).
- Teaching Strategies and Technology Tools**
Students are expected to use **Moodle**, and only **basic scientific calculators** are allowed (Graphic and programmable calculators are **NOT** allowed).

X. UNIVERSITY RULES AND REGULATIONS

- **General Academic Information**

[Undergraduate Academic Information](#) , [Graduate Academic Information](#)

- **University - Course Withdrawal Policy**

A student can withdraw from only one required course per semester. Students who wish to withdraw from more than one required course in any given semester must petition the appropriate faculty committee for permission to do so.

Check Withdrawal Information from [undergraduate](#) and [graduate](#) catalogue.

- **Academic Integrity (cheating and plagiarism):**

Please refer to [AUB Student Code of Conduct](#), in particular section 1.1, which concerns academic misconduct including cheating, plagiarism, in-class disruption, and dishonesty. Please be aware that misconduct is vigorously prosecuted and that AUB has a zero-tolerance policy. Course policy is that credible evidence of cheating will result in course failure.

- **Accessibility Statement to Acknowledge the Unique Learning Needs of Students with Disabilities:**

AUB strives to make learning experiences as accessible as possible. If you anticipate or experience academic barriers due to a diagnosed disability (including learning disability, mental health, chronic or temporary medical conditions), please contact the [Accessible Education Office](#) as soon as possible in order to help establish reasonable accommodations and facilitate a smooth learning process: accessibility@aub.edu.lb; +961-1350000, x3151; West Hall, 338.

- **[Non-Discrimination and Title IX Statement](#)**

- AUB is committed to the principle of equal opportunity in admission, education, and employment. Consistent with this commitment and with Lebanese, U.S., and other applicable laws, it is the policy of AUB not to tolerate discrimination or discriminatory harassment, including sex discrimination and sexual harassment misconduct (as described in the relevant university policies) and to provide those who feel they are victims of discriminatory harassment (including sexual harassment) with mechanisms for seeking redress, as described in the applicable procedures.

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In alignment with the objectives of Title IX of the US Education Amendments of 1972, which prohibits sex discrimination in education and interprets sex discrimination as including sexual harassment, AUB maintains a Title IX Coordinator in the Office of the President. The office is also supported by a network of faculty and staff members including an Advisory Council, Title IX Deputies, and qualified Panel Members for formal investigations. Members of the network are certified by the Association of Title IX Administrators (ATIXA).

- Inquiries regarding any kind of discrimination or discriminatory harassment may be directed to the Title IX Coordinator, Mitra Tauk: AUB | Office of the President | College Hall | Room 425 | 01-350000 ext.: 2514, titleix@aub.edu.lb

You also have the right to raise concerns to the US Department of Education's Office for Civil Rights that oversees compliance with Title IX.

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XI. AUB GRADING SYSTEM

<https://www.aub.edu.lb/Registrar/Pages/academic-information.aspx>

Numeric Course Grade to Letter Course Grade, Grade Conversion Chart

Starting with Numeric Course Grade /100	Corresponding Course Letter Grade
< 60	F
60	D
61–62	D+
63–65	C–
66–68	C
69–71	C+
72–74	B–
75–78	B
79–82	B+
83–86	A–
87–92	A
93–100	A+
Course Letter Grade	Quality Points
A+	4.3
A	4.0
A–	3.7
B+	3.3
B	3.0
B–	2.7
C+	2.3
C	2.0
C–	1.7
D+	1.3
D	1.0
F	0.0

Note: The GPA at AUB is capped at 4.0

I	P	PR	W	NP
Incomplete	Pass	In Progress	Withdraw	No Pass